

Construction of localized solutions to a generalized (3 + 1)-dimensional Burgers equation

Xiaorui Hu^{*,a}, Rui Zhang^a, Shoufeng Shen^a, Wenxiu Ma^b

^a Department of Applied Mathematics, Zhejiang University of Technology, Hangzhou 310023, China

^b Department of Mathematics and Statistics, University of South Florida, Tampa 33620-5700, USA

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ABSTRACT

Finding exact and analytic solutions of nonlinear system in high dimensions is a difficult but meaningful work. In this paper, by means of the symbolic package Maple, we investigate a Painlevé integrable (3 + 1)-dimensional generalized Burgers (gBurgers) equation. Starting from the Cole-Hopf transformation with different seed solutions, abundant localized solutions are provided, including new variable separation solutions, lumps, lump-“multiple-soliton” solutions and other interaction solutions. Specifically, the lump-two soliton solution and lump-soliton-periodic solution are depicted by the three-dimensional plots and contour plots at different times, respectively. The methods and all the resulting solutions presented in this paper can be generalized to the Painlevé integrable (N + 1)-dimensional Burgers equation.

1. Introduction

The (1 + 1)-dimensional Burgers equation

$$u_t = uu_x + \alpha u_{xx} \quad (1)$$

has been widely used in fluid mechanics, traffic flow's model, heat conduction and shock waves. Eq. (1) can be generalized to (2 + 1) dimensions, saying

$$\begin{aligned} u_t &= \gamma uu_y + \alpha v u_x + \lambda (\alpha u_{xx} + \gamma u_{yy}), \\ u_x &= v_y. \end{aligned} \quad (2)$$

Exact solutions of this equation such as the multiple kink solution, rational function solution and variable separation solution have been obtained [1–4]. By means of the Painlevé analysis, Wazwaz [5] has proposed the most generalized Painlevé integrable (3 + 1)-dimensional Burgers system (gBurgers) with four parameters:

$$\begin{aligned} u_t &= \gamma uu_y + \alpha v u_x + \beta w u_z + \lambda (\alpha u_{xx} + \gamma u_{yy} + \beta u_{zz}), \\ u_x &= v_y, \\ u_z &= w_y. \end{aligned} \quad (3)$$

Eq. (3) will be reduced back to Eq. (2) for z-independent and Eq. (1) for both z-independent and y-independent. For the special case $\gamma = \alpha = \beta = 2$, $\lambda = \frac{1}{2}$, the Painlevé property, Bäcklund transformation and rogue wave solution of Eq. (3) have been derived in Refs. [6–9].

* Corresponding author.

E-mail address: lansexiaoer@163.com (X. Hu).

In the same way, the Painlevé integrable $(N + 1)$ -dimensional Burgers system which is extended naturally from Eq. (3) is in the form [9,10]

$$\begin{cases} u_{1,t} = \sum_{j=1}^N \theta_j u_j u_{1,x_j} + \lambda \sum_{j=1}^N \theta_j u_{1,x_j x_j}, \\ u_{1,x_i} = u_{i,x_1}, \quad i = 2, \dots, N \end{cases} \quad (4)$$

with θ_j and λ being arbitrary constants. Notice that Eq. (4) is different from the following classical multi-dimensional Burgers system [11,12]

$$u_{i,t} = \sum_{j=1}^N \theta_j u_j u_{i,x_j} + \lambda \sum_{j=1}^N \theta_j u_{i,x_j x_j}, \quad i = 1, 2, \dots, N, \quad (5)$$

where $u_1, u_2, \dots, u_N$ are the fluid velocity fields. The Burgers Eq. (5) has found interesting applications [11] in cosmology, condensed matter, statistical physics and even in non-physics problems such as vehicular traffic, etc. Different from Eq. (4), Eq. (5) does not possess Painlevé property [13]. However, we verified that both two types of Burgers Eqs. (4) and (5) can be linearized into one heat equation

$$f_t = \lambda \sum_{j=1}^N \theta_j f_{x_j x_j} \quad (6)$$

under an N -dimensional Cole-Hopf transformation [12,14,15]

$$u_i = 2\lambda (\ln f)_{x_i}, \quad i = 1, 2, \dots, N. \quad (7)$$

During the past two decades, the multilinear variable separation approach (MLVSA)[16,17] has been developed well for many nonlinear systems. The key to this approach is the lower-dimensional arbitrary function in the “seed solution” and the transformation of this kind of arbitrariness. Besides, upon taking the “seed solution” as constants, one can derive abundant lumps and the interaction solutions between lumps and another kind of exact solutions to nonlinear partial differential equations (NPDEs) in $(2 + 1)$ -dimensions [18–29]. Nevertheless, in the $(3 + 1)$ -dimensional case, only lump-type solutions [30–33] are presented for the NPDEs, such as Jimbo-Miwa equation and Jimbo-Miwa like equation. The lumps are localized in all directions in space while the lump-type solutions are rationally localized in almost all but not all directions in space. In the recent paper [34], the authors provide lump-type (not lump) solutions, lump-one kink solution and lump-two kinks solution for Eq. (3) with $\lambda = \frac{1}{2}$.

In this paper, we will consider more localized solutions for the $(3 + 1)$ -dimensional gBurgers Eq. (3), including variable separation solutions, real lump solution, lump-“multiple-soliton” solutions and other interaction solutions (such as lump-soliton-periodic solution). It shows, specifically, the methods and all the resulting solutions mentioned in this paper can be generalized to $(N + 1)$ -dimensional Burgers Eq. (4). For the classical multi-dimensional Burgers system (5), we can not use the MLVSA to obtain the corresponding variable separation solutions which are shown in Section 2. However, the lump and the related interaction solutions presented in Section 3 also exist for Eq. (5).

2. New variable separation solutions of gBurgers Eq. (3)

For Eq. (3), consider the following Cole-Hopf transformation [10,14,15]

$$u = 2\lambda (\ln f)_y + u^*, \quad v = 2\lambda (\ln f)_x + v^*, \quad w = 2\lambda (\ln f)_z + w^*, \quad (8)$$

where $\{u^* \equiv u^*(x, y, z, t), v^* \equiv v^*(x, y, z, t), w^* \equiv w^*(x, y, z, t)\}$ is defined as the seed solution of Eq. (3).

If the seed solution is fixed as

$$u^* = 0, \quad v^* = \bar{v}(x, z, t), \quad w^* = \bar{w}(x, z, t), \quad (9)$$

we have

$$\begin{aligned} (\ln f)_{yt} = & 4\lambda \gamma (\ln f)_y (\ln f)_{yy} + \alpha [2\lambda (\ln f)_x + \bar{v}] (\ln f)_{xy} + \beta [2\lambda (\ln f)_z + \bar{w}] (\ln f)_{zy} \\ & + \lambda [\alpha (\ln f)_{xxy} + \gamma (\ln f)_{yyy} + \beta (\ln f)_{yzz}] \end{aligned} \quad (10)$$

with $\bar{v} \equiv \bar{v}(x, z, t)$ and $\bar{w} \equiv \bar{w}(x, z, t)$. Integrate the above equation about y , we can obtain

$$f_t = \alpha \bar{v} f_x + \beta \bar{w} f_z + \lambda (\alpha f_{xx} + \gamma f_{yy} + \beta f_{zz}). \quad (11)$$

Adopt a multi-linear variable separation ansatz as

$$f = q_1(y, t) + \sum_{j=2}^M q_j(y, t) p_j(x, z, t), \quad 2 \leq M \leq 3. \quad (12)$$

Eq. (11) becomes

$$\lambda\gamma q_{1yy} - q_{1t} + \sum_{j=2}^M p_j(\lambda\gamma q_{jyy} - q_{jt}) + \sum_{j=2}^M q_j(\alpha\tilde{v}p_{jx} + \beta\tilde{w}p_{jz} + \lambda\alpha p_{jxx} + \lambda\beta p_{jzz} - p_{jt}) = 0 \quad (13)$$

with $q_j \equiv q_j(y, t)$ and $p_j \equiv p_j(x, z, t)$. Then one can choose

$$\lambda\gamma q_{1yy} - q_{1t} = 0, \quad i = 1, 2, \dots, M, \quad (14)$$

$$\alpha\tilde{v}p_{jx} + \beta\tilde{w}p_{jz} + \lambda\alpha p_{jxx} + \lambda\beta p_{jzz} - p_{jt} = 0, \quad j = 2, \dots, M. \quad (15)$$

By solving the above equations, we can obtain the following two results.

For $M = 2$ in (12), the general exact solution (8) is shown as:

$$\begin{cases} u = 2\lambda \frac{q_{1y} + q_{2y}p_2}{q_1 + q_2p_2}, \\ v = 2\lambda \frac{q_2p_{2x}}{q_1 + q_2p_2} + \frac{p_{2t} - \lambda\alpha p_{2xx} - \lambda\beta p_{2zz} - \beta\tilde{w}p_{2z}}{\alpha p_{2x}}, \\ w = 2\lambda \frac{q_2p_{2z}}{q_1 + q_2p_2} + \tilde{w}, \end{cases} \quad (16)$$

where $\{q_1, q_2\}$ satisfy the linear differential equations (14) and $\{p_2, \tilde{w}\}$ are arbitrary.

For $M = 3$ in (12), the general exact solution (8) is shown as:

$$\begin{cases} u = 2\lambda \frac{q_{1y} + q_{2y}p_2 + q_{3y}p_3}{q_1 + q_2p_2 + q_3p_3}, \\ v = 2\lambda \frac{q_2p_{2x} + q_3p_{3x}}{q_1 + q_2p_2 + q_3p_3} + \frac{\begin{vmatrix} p_{2t} - \lambda\alpha p_{2xx} - \lambda\beta p_{2zz} & p_{2z} \\ p_{3t} - \lambda\alpha p_{3xx} - \lambda\beta p_{3zz} & p_{3z} \end{vmatrix}}{\alpha \begin{vmatrix} p_{2x} & p_{2z} \\ p_{3x} & p_{3z} \end{vmatrix}}, \\ w = 2\lambda \frac{q_2p_{2z} + q_3p_{3z}}{q_1 + q_2p_2 + q_3p_3} + \frac{\begin{vmatrix} p_{2x} & p_{2t} - \lambda\alpha p_{2xx} - \lambda\beta p_{2zz} \\ p_{3x} & p_{3t} - \lambda\alpha p_{3xx} - \lambda\beta p_{3zz} \end{vmatrix}}{\beta \begin{vmatrix} p_{2x} & p_{2z} \\ p_{3x} & p_{3z} \end{vmatrix}}, \end{cases} \quad (17)$$

where $\{q_1, q_2, q_3\}$ satisfy the heat conductive equations (14) and $\{p_2, p_3\}$ are arbitrary.

By selecting appropriate p_j ($j = 2, 3$) and q_j ($j = 1, 2, 3$), one can construct abundant localized coherent structures for the potential u_x and u_z , such as the solitoffs, dromions, lumps, breathers, instantons, peakons, compactons, localized chaotic and fractal patterns, and so on. Here we will not dwell on it.

3. Lump and abundant interaction solutions of gBurgers Eq. (3)

For the C-integrable nonlinear Eq. (3), the transformation [14,15,34]

$$u = 2\lambda(\ln f)_y + u_0, \quad v = 2\lambda(\ln f)_x + v_0, \quad w = 2\lambda(\ln f)_z + w_0 \quad (18)$$

with real constants $\{u_0, v_0, w_0\}$ leads it to the following linear differential equation:

$$f_t = u_0\gamma f_y + v_0\alpha f_x + w_0\beta f_z + \lambda(\alpha f_{xx} + \gamma f_{yy} + \beta f_{zz}). \quad (19)$$

To search for interaction solutions of Eq. (3), we consider

$$f = f_0 + \xi^2 + \sum_{j=1}^n \phi_j(\eta_j), \quad (\xi^2 = \vec{\xi} \cdot \vec{\xi} = \sum_{i=1}^m \xi_i^2). \quad (20)$$

Here, ϕ_j is a function of η_j with $\eta_j = a_jx + b_jy + c_jz + d_jt + e_j$. $\vec{\xi}$ is an m component real vector related to the constant vectors $\vec{k} = (k_1, k_2, \dots, k_m)$, $\vec{h} = (h_1, h_2, \dots, h_m)$, $\vec{l} = (l_1, l_2, \dots, l_m)$, $\vec{\omega} = (\omega_1, \omega_2, \dots, \omega_m)$ and $\vec{s} = (s_1, s_2, \dots, s_m)$ by

$$\vec{\xi} = \vec{k}x + \vec{h}y + \vec{l}z + \vec{\omega}t + \vec{s}. \quad (21)$$

Some special cases of (20) with $m = 3, n = 1$ and $m = 4, n = 1$ are used to obtain lumps and their interaction solutions to (3+1)-dimensional linear PDEs [35] and (4+1)-dimensional linear PDEs [36], respectively. After substituting (20) into (19), we have:

$$A_5 + A_1x + A_2y + A_3z + A_4t + \sum_{j=1}^n (a_j\alpha v_0 + b_j\gamma u_0 + c_j\beta w_0 - d_j) \frac{d\phi_j}{d\eta_j} + \sum_{j=1}^n \lambda(\alpha a_j^2 + \gamma b_j^2 + \beta c_j^2) \frac{d^2\phi_j}{d\eta_j^2} = 0 \quad (22)$$

with

$$\begin{cases} A_1 \equiv \gamma u_0(\vec{k} \cdot \vec{h}) + \alpha v_0(\vec{k} \cdot \vec{k}) + \beta w_0(\vec{k} \cdot \vec{l}) - \vec{k} \cdot \vec{\omega}, \\ A_2 \equiv \gamma u_0(\vec{h} \cdot \vec{h}) + \alpha v_0(\vec{h} \cdot \vec{k}) + \beta w_0(\vec{h} \cdot \vec{l}) - \vec{h} \cdot \vec{\omega}, \\ A_3 \equiv \gamma u_0(\vec{l} \cdot \vec{h}) + \alpha v_0(\vec{l} \cdot \vec{k}) + \beta w_0(\vec{l} \cdot \vec{l}) - \vec{l} \cdot \vec{\omega}, \\ A_4 \equiv \gamma u_0(\vec{\omega} \cdot \vec{h}) + \alpha v_0(\vec{\omega} \cdot \vec{k}) + \beta w_0(\vec{\omega} \cdot \vec{l}) - \vec{\omega} \cdot \vec{\omega}, \\ A_5 \equiv \gamma u_0(\vec{s} \cdot \vec{h}) + \alpha v_0(\vec{s} \cdot \vec{k}) + \beta w_0(\vec{s} \cdot \vec{l}) + \lambda(\gamma \vec{h} \cdot \vec{h} + \alpha \vec{k} \cdot \vec{k} + \beta \vec{l} \cdot \vec{l}) - \vec{s} \cdot \vec{\omega}. \end{cases} \quad (23)$$

Starting from Eq. (22), we focus on the following three cases.

Case 1. Lump solution ($m \geq 3$)

Take $\phi_j = 0$ in Eq. (22). Now solution (20) with $\phi_j = 0$ is a pure algebraic solution which may be a lump (a rogue wave) if it is localized in space only (in both space and time). Direct symbolic computations with Maple to solve $A_i = 0$ ($i = 1, \dots, 5$) can determine $m + 1$ constraining equations

$$\begin{cases} \vec{\omega} = \vec{k} \alpha v_0 + \vec{h} \gamma u_0 + \vec{l} \beta w_0, \\ \vec{k}^2 \alpha + \vec{h}^2 \gamma + \vec{l}^2 \beta = 0 \end{cases} \quad (24)$$

with $\vec{k}^2 = \vec{k} \cdot \vec{k}$, $\vec{h}^2 = \vec{h} \cdot \vec{h}$, $\vec{l}^2 = \vec{l} \cdot \vec{l}$. Eq. (20) becomes

$$f = f_0 + \sum_{i=1}^m (k_i x + h_i y + l_i z + \omega_i t + s_i)^2, \quad (25)$$

and leads to a rational function solution of gBurgers equation:

$$\begin{cases} u = 4\lambda \frac{h_1 \xi_1 + h_2 \xi_2 + \dots + h_m \xi_m}{f_0 + \xi_1^2 + \xi_2^2 + \dots + \xi_m^2} + u_0, \\ v = 4\lambda \frac{k_1 \xi_1 + k_2 \xi_2 + \dots + k_m \xi_m}{f_0 + \xi_1^2 + \xi_2^2 + \dots + \xi_m^2} + v_0, \\ w = 4\lambda \frac{l_1 \xi_1 + l_2 \xi_2 + \dots + l_m \xi_m}{f_0 + \xi_1^2 + \xi_2^2 + \dots + \xi_m^2} + w_0, \end{cases} \quad (26)$$

where

$$\begin{cases} \xi_i = k_i x + h_i y + l_i z + (k_i \alpha v_0 + h_i \gamma u_0 + l_i \beta w_0)t + s_i, \quad i = 1, 2, \dots, m, \\ (\sum_{i=1}^m k_i^2) \alpha + (\sum_{i=1}^m h_i^2) \gamma + (\sum_{i=1}^m l_i^2) \beta = 0. \end{cases} \quad (27)$$

Due to the dispersion relation in (24), the solution (26) is only a pure lump solution but not a rogue wave (instanton) solution for $m \geq 3$. To guarantee f being positive, we can let $f_0 > 0$. For example, by adopting $m = 3$ and

$$\begin{vmatrix} k_1 & h_1 & l_1 \\ k_2 & h_2 & l_2 \\ k_3 & h_3 & l_3 \end{vmatrix} \neq 0, \quad (28)$$

we have

$$\lim_{x^2+y^2+z^2 \rightarrow \infty} u(x, y, z, t) = u_0, \quad \lim_{x^2+y^2+z^2 \rightarrow \infty} v(x, y, z, t) = v_0, \quad \lim_{x^2+y^2+z^2 \rightarrow \infty} w(x, y, z, t) = w_0 \quad (29)$$

at any given time t . Hence, we remark that the presented solution (26) is a lump solution, rationally localized in all directions in the (x, y, z) -space. For the following $(2+1)$ -dimensional reduction

$$z = \mu_1 x + \mu_2 y + \mu_3 t + \mu_0, \quad (30)$$

the lump solution becomes a real bright-dark lump on the (x, y) -plane [26]. Various studies show the existence of lumps to nonlinear PDEs in $(2+1)$ -dimensions. Nevertheless, in the $(3+1)$ -dimensional case, only lump-type solutions are presented for nonlinear PDEs, which are rationally localized in almost all but not all directions in space. All presented exact solutions to the $(3+1)$ -dimensional Jimbo-Miwa equation [31,32], $(3+1)$ -dimensional Jimbo-Miwa like equation [33] and $(3+1)$ -dimensional Burgers equation [34] are not rationally localized in all directions in space.

Case 2. Interactions between lump and multiple-soliton solutions

In this case, we adopt $\phi_j = C_j e^{\eta_j}$ with C_j being arbitrary real constant. Vanishing all the coefficients of e^{η_j} and x, y, z, t in Eq. (22) leads to

$$\begin{cases} \vec{\omega} = \vec{k} \alpha v_0 + \vec{h} \gamma u_0 + \vec{l} \beta w_0, \\ d_j = a_j \alpha v_0 + b_j \gamma u_0 + c_j \beta w_0 + \lambda(\alpha a_j^2 + \gamma b_j^2 + \beta c_j^2), \quad j = 1, 2, \dots, n, \\ \vec{k}^2 \alpha + \vec{h}^2 \gamma + \vec{l}^2 \beta = 0. \end{cases} \quad (31)$$

That is to say if the form of f satisfies

$$f = f_0 + \sum_{i=1}^m (k_i x + h_i y + l_i z + \omega_i t + s_i)^2 + \sum_{j=1}^n C_j e^{a_j x + b_j y + c_j z + d_j t}, \quad m \geq 3 \quad (32)$$

with the conditions (31), then (18) is a lump-multiple soliton interaction solution to the gBurgers equation:

$$\begin{cases} u = 2\lambda \frac{\sum_{i=1}^m 2h_i \xi_i + \sum_{j=1}^n C_j b_j e^{\eta_j}}{f_0 + \sum_{i=1}^m \xi_i^2 + \sum_{j=1}^n C_j e^{\eta_j}} + u_0, \\ v = 2\lambda \frac{\sum_{i=1}^m 2k_i \xi_i + \sum_{j=1}^n C_j a_j e^{\eta_j}}{f_0 + \sum_{i=1}^m \xi_i^2 + \sum_{j=1}^n C_j e^{\eta_j}} + v_0, \\ w = 2\lambda \frac{\sum_{i=1}^m 2l_i \xi_i + \sum_{j=1}^n C_j c_j e^{\eta_j}}{f_0 + \sum_{i=1}^m \xi_i^2 + \sum_{j=1}^n C_j e^{\eta_j}} + w_0, \end{cases} \quad (33)$$

where

$$\begin{cases} \xi_i = k_i x + h_i y + l_i z + (k_i \alpha v_0 + h_i \gamma u_0 + l_i \beta w_0) t + s_i, \quad i = 1, 2, \dots, m, \\ \eta_j = a_j x + b_j y + c_j z + [a_j \alpha v_0 + b_j \gamma u_0 + c_j \beta w_0 + \lambda(a_j^2 \alpha + b_j^2 \gamma + c_j^2 \beta)] t, \quad j = 1, 2, \dots, n, \\ (\sum_{i=1}^m k_i^2) \alpha + (\sum_{i=1}^m h_i^2) \gamma + (\sum_{i=1}^m l_i^2) \beta = 0. \end{cases} \quad (34)$$

To see this lump-multiple soliton interaction solution clearly, we take $m = 3$ and $n = 1$ in (33) and (34) for a concrete example. Let

$$f_0 > 0, \quad C_1 > 0, \quad (35)$$

and

$$\begin{vmatrix} k_1 & h_1 & l_1 & k_1 \alpha v_0 + h_1 \gamma u_0 + l_1 \beta w_0 \\ k_2 & h_2 & l_2 & k_2 \alpha v_0 + h_2 \gamma u_0 + l_2 \beta w_0 \\ k_3 & h_3 & l_3 & k_3 \alpha v_0 + h_3 \gamma u_0 + l_3 \beta w_0 \\ a_1 & b_1 & c_1 & a_1 \alpha v_0 + b_1 \gamma u_0 + c_1 \beta w_0 + \lambda(a_1^2 \alpha + b_1^2 \gamma + c_1^2 \beta) \end{vmatrix} \neq 0. \quad (36)$$

Solution (33) indicates a lump-one soliton interaction. Through analyzing (33) with (35) and (36), we can see

$$\begin{cases} \lim_{\eta_1 \rightarrow +\infty} u(x, y, z, t) = 2\lambda b_1 + u_0, \quad \lim_{\eta_1 \rightarrow +\infty} v(x, y, z, t) = 2\lambda a_1 + v_0, \quad \lim_{\eta_1 \rightarrow +\infty} w(x, y, z, t) = 2\lambda c_1 + w_0, \\ \lim_{\eta_1 \rightarrow -\infty} u(x, y, z, t) = u_0, \quad \lim_{\eta_1 \rightarrow -\infty} v(x, y, z, t) = v_0, \quad \lim_{\eta_1 \rightarrow -\infty} w(x, y, z, t) = w_0. \end{cases} \quad (37)$$

When $\eta_1 \rightarrow +\infty$ in (33), the algebraic quadratic function part can be neglected. But for the region $\eta_1 < 0$, the algebraic part will play a dominant role. In this lump-one soliton interaction solution, the lump exists only in half space-time for $\eta_1 < 0$. In other words, the lump is cut off by one exponentially plane soliton. For any (2 + 1)-dimensional reduction given by (30), this interaction solution has a fusion or fission phenomena among a real bright-dark lump and one-kink solution.

Similarly, the above analysis also works for the lump-multiple soliton interaction solutions. Specifically, by taking $p = \frac{1}{2}$, $\mu = 2$, $\beta = 2$, $\alpha = -\frac{50}{3}$, $u_0 = 1$ and

$$f = 2 + (x + 3y + z + 6t + 1)^2 + (x - y + z - 2t + 1)^2 + (x + 2y + 3z + 4t)^2 + e^{2x-y+z-\frac{100}{3}t} + e^{3x+2y+z-66t}, \quad (38)$$

we depict a lump-two soliton interaction solution with the three-dimensional plots and corresponding contour plots at different times for u in Fig. 1.

Case 3. Interactions between lump and other waves

Rather than adopting $\phi_j = 0$ and $\phi_j = C_j e^{\eta_j}$, we set all coefficients of $\frac{d^2 \phi_j}{d\eta_j^2}$, $\frac{d\phi_j}{d\eta_j}$ and x, y, z, t in (22) to be zeros. Then a system of equations are obtained:

$$\begin{cases} \vec{\omega} = \vec{k} \alpha v_0 + \vec{h} \gamma u_0 + \vec{l} \beta w_0, \\ d_j = a_j \alpha v_0 + b_j \gamma u_0 + c_j \beta w_0, \quad j = 1, 2, \dots, n, \\ a_j^2 \alpha + b_j^2 \gamma + c_j^2 \beta = 0, \quad j = 1, 2, \dots, n, \\ \vec{k}^2 \alpha + \vec{h}^2 \gamma + \vec{l}^2 \beta = 0. \end{cases} \quad (39)$$

Thus if we take

$$\begin{aligned} f &= f_0 + \sum_{i=1}^m \xi_i^2 + \sum_{j=1}^n \phi_j(\eta_j), \\ (\xi_i &= k_i x + h_i y + l_i z + \omega_i t + s_i, \quad \eta_j = a_j x + b_j y + c_j z + d_j t,) \end{aligned} \quad (40)$$

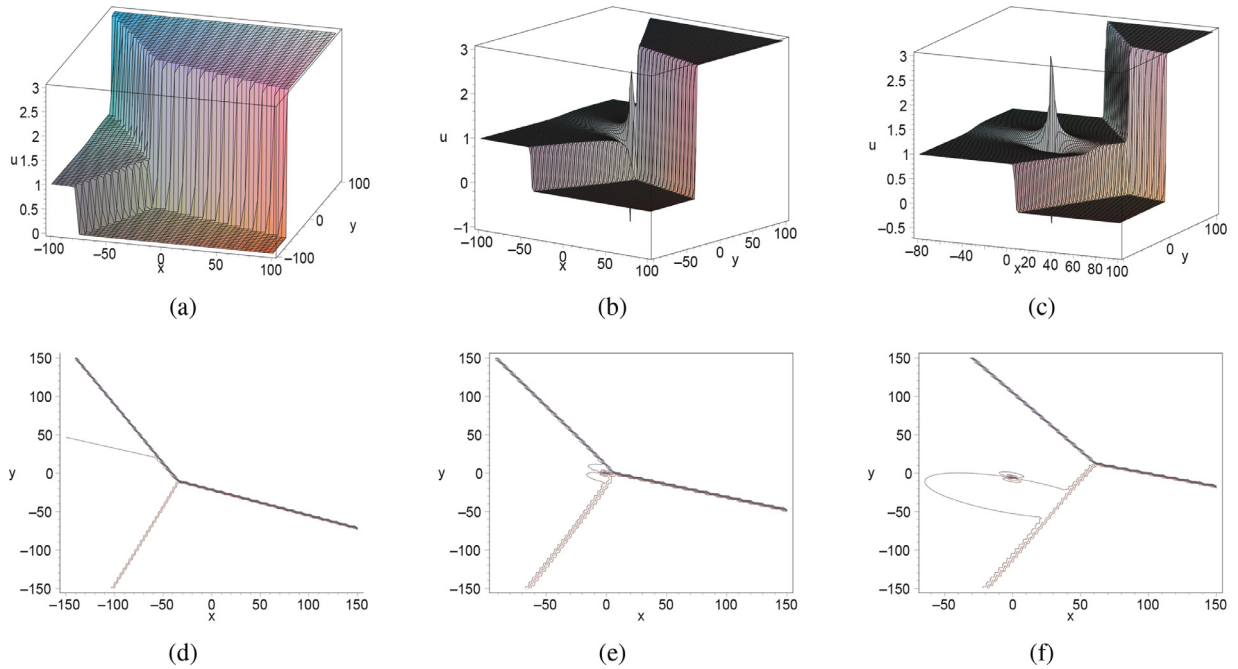


Fig. 1. The lump-two soliton interaction solution for u when $t = -3, 0, 3$ and $z = 0$: 3d plots (a–c) and contour plots (d–f).

where the parameters involved satisfy Eqs. (39) and every ϕ_j is arbitrary real function of η_j , then the gBurgers equation has an exact solution

$$\begin{cases} u = 2\lambda \frac{\sum_{i=1}^m 2h_i \xi_i + \sum_{j=1}^n b_j \phi_j'(\eta_j)}{f_0 + \sum_{i=1}^m \xi_i^2 + \sum_{j=1}^n \phi_j(\eta_j)} + u_0, \\ v = 2\lambda \frac{\sum_{i=1}^m 2k_i \xi_i + \sum_{j=1}^n a_j \phi_j'(\eta_j)}{f_0 + \sum_{i=1}^m \xi_i^2 + \sum_{j=1}^n \phi_j(\eta_j)} + v_0, \\ w = 2\lambda \frac{\sum_{i=1}^m 2l_i \xi_i + \sum_{j=1}^n c_j \phi_j'(\eta_j)}{f_0 + \sum_{i=1}^m \xi_i^2 + \sum_{j=1}^n \phi_j(\eta_j)} + w_0. \end{cases} \quad (41)$$

It is seen that different choices of ϕ_j in (41) contribute rich interaction solutions between lump and other travelling waves. A fact that we need remark is all the resulting solutions satisfy a determinant identity

$$\begin{vmatrix} k_{i_1} & h_{i_1} & l_{i_1} & w_{i_1} \\ k_{i_2} & h_{i_2} & l_{i_2} & w_{i_2} \\ k_{i_3} & h_{i_3} & l_{i_3} & w_{i_3} \\ a_j & b_j & c_j & d_j \end{vmatrix} = 0, \quad (42)$$

where $1 \leq i_1 < i_2 < i_3 \leq m$ and $1 \leq j \leq n$ are arbitrary natural numbers.

Specifically, we take $m = 3$ and $n = 2$ with

$$\phi_1 = B_1 \cosh(\eta_1), \quad \phi_2 = B_2 \cos(\eta_2) \quad (\text{or} \quad \phi_2 = B_3 \sin(\eta_2)) \quad (43)$$

to fulfill the positivity of f . The resulting solution is a lump-soliton-periodic wave solution:

$$\begin{cases} u = 2\lambda \frac{2(h_1 \xi_1 + h_2 \xi_2 + h_3 \xi_3) + B_1 b_1 \sinh(\eta_1) - B_2 b_2 \sin(\eta_2)}{f_0 + \xi_1^2 + \xi_2^2 + \xi_3^2 + B_1 \cosh(\eta_1) + B_2 \cos(\eta_2)} + u_0, \\ v = 2\lambda \frac{2(k_1 \xi_1 + k_2 \xi_2 + k_3 \xi_3) + B_1 a_1 \sinh(\eta_1) - B_2 a_2 \sin(\eta_2)}{f_0 + \xi_1^2 + \xi_2^2 + \xi_3^2 + B_1 \cosh(\eta_1) + B_2 \cos(\eta_2)} + v_0, \\ w = 2\lambda \frac{2(l_1 \xi_1 + l_2 \xi_2 + l_3 \xi_3) + B_1 c_1 \sinh(\eta_1) - B_2 c_2 \sin(\eta_2)}{f_0 + \xi_1^2 + \xi_2^2 + \xi_3^2 + B_1 \cosh(\eta_1) + B_2 \cos(\eta_2)} + w_0 \end{cases} \quad (44)$$

with

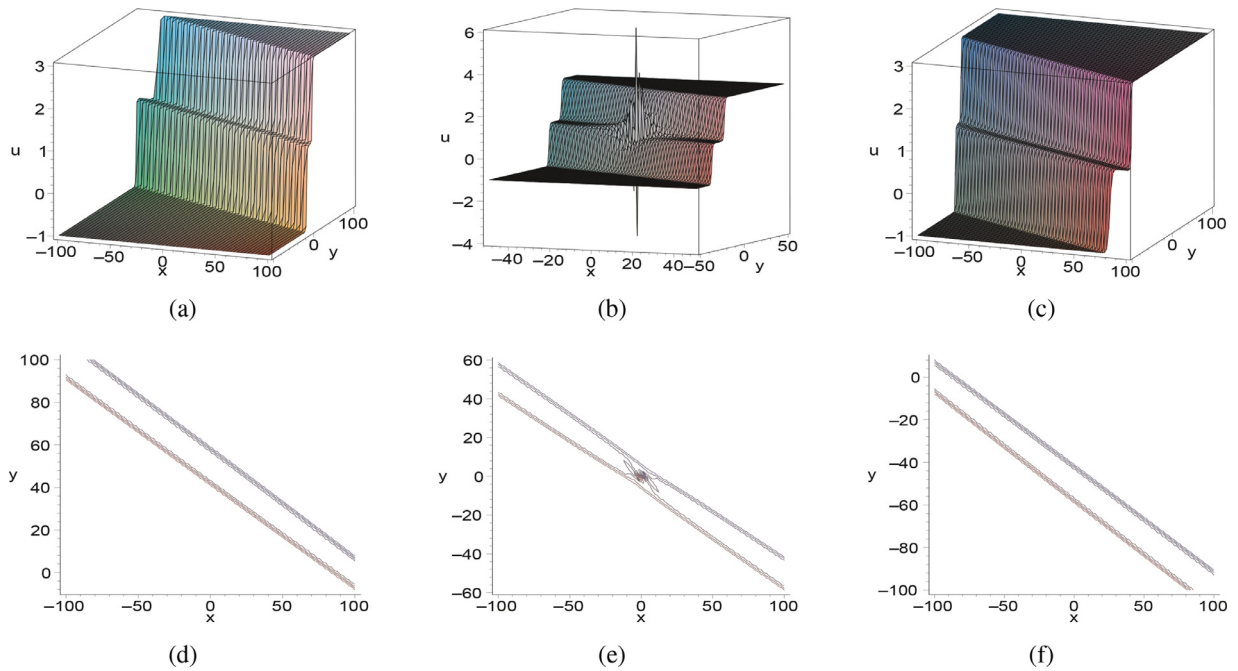


Fig. 2. The lump-soliton-periodic solution for u when $t = -50, 0, 50$ and $z = 0$: 3d plots (a–c) and contour plots (d–f).

$$\begin{cases} \xi_i = k_i x + h_i y + l_i z + (k_i \alpha v_0 + h_i \gamma u_0 + l_i \beta w_0) t + s_i, & i = 1, 2, 3, \\ \eta_j = a_j x + b_j y + c_j z + (a_j \alpha v_0 + b_j \gamma u_0 + c_j \beta w_0) t, & j = 1, 2, \\ a_j^2 \alpha + b_j^2 \gamma + c_j^2 \beta = 0, & j = 1, 2, \\ (\sum_{i=1}^3 k_i^2) \alpha + (\sum_{i=1}^3 h_i^2) \gamma + (\sum_{i=1}^3 l_i^2) \beta = 0. \end{cases} \quad (45)$$

Due to the identity (42), the solution (44) is not a rogue wave for the (3+1)-dimensional gBurgers Eq. (3). However, for $z = \text{constant}$, the resulting solution (44) only with (x, y) can generate a rogue wave. For example, we take $p = \frac{1}{2}$, $\mu = -1$, $\beta = 2$, $\alpha = 2$, $u_0 = 1$ and

$$\begin{aligned} f = & 10 + (x + y + z + 3t)^2 + (x - y + 3t)^2 + (x + \sqrt{14}y + 2z + (6 - \sqrt{14})t)^2 + 0.01 \cosh(x + 2y + z + 2t) \\ & - 10 \cos(2x + \sqrt{26}y + 3z + (10 - \sqrt{26})t). \end{aligned} \quad (46)$$

The corresponding three-dimensional plots and contour plots are presented in Fig. 2. Through the figure, one can see that the lump-periodic wave only exists for a small period of time in this lump-soliton-periodic solution, which reflect a dynamical behavior of the rogue wave [37–39].

4. Conclusions

In this paper, our interest has been focused on a Painlevé integrable (3+1)-dimensional gBurgers equation with four model parameters. Abundant exact solutions were successfully constructed by adopting two different types of “seed solution”. For the first seed solution (9), a multi-linear variable separation ansatz was used to produce new general exact solutions (16) and (17) which contained two arbitrary functions. For the second seed solution (18), three different sets of exact solutions were determined respectively. Case one offered the lump solution which decayed in all space directions rather than the lump-type solution which decayed in almost all but not all directions in space. Case two provided the lump-multiple-soliton solutions, which would produce the fissions or fusions for the lump being cut off by the multiple-soliton. Specifically, we drew a lump-two soliton solution at different times in Fig. 1. Case three demonstrated abundant interaction solutions between lump and other traveling waves, in which we laid stress on a lump-soliton-periodic solution and then clearly depicted it in Fig. 2.

In addition, all the results obtained in this paper can be generalized to the $(N + 1)$ -dimensional gBurgers system (4). Similar to Section 2, here we present a new variable separation solution for Burgers Eq. (4) in $N + 1$ dimensions:

$$\begin{cases} u_1 = 2\lambda(\ln f)_{x_1} + 0, \\ u_i = 2\lambda(\ln f)_{x_i} + \tilde{u}_i(x_2, x_3, \dots, x_N), & i = 2, 3, \dots, M, \\ u_j = 2\lambda(\ln f)_{x_j} + \tilde{u}_j(x_2, x_3, \dots, x_N), & j = M + 1, \dots, N \end{cases} \quad (47)$$

with

$$f = q_1(x_1, t) + \sum_{j=2}^M q_j(x_1, t) p_j(x_2, x_3, \dots, x_N), \quad 2 \leq M \leq N, \quad (48)$$

$$\tilde{u}_i = \frac{\begin{vmatrix} p_{2x_2} & \cdots & p_{2x_{i-1}} & p_{2t} - \lambda \sum_{l=2}^N \theta_l p_{2x_l x_l} - \sum_{k=M+1}^N \theta_k p_{2x_k} \tilde{u}_k & p_{2x_{i+1}} & \cdots & p_{2x_M} \\ p_{3x_2} & \cdots & p_{3x_{i-1}} & p_{3t} - \lambda \sum_{l=2}^N \theta_l p_{3x_l x_l} - \sum_{k=M+1}^N \theta_k p_{3x_k} \tilde{u}_k & p_{3x_{i+1}} & \cdots & p_{3x_M} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ p_{Mx_2} & \cdots & p_{Mx_{i-1}} & p_{Mt} - \lambda \sum_{l=2}^N \theta_l p_{Mx_l x_l} - \sum_{k=M+1}^N \theta_k p_{Mx_k} \tilde{u}_k & p_{Mx_{i+1}} & \cdots & p_{Mx_M} \end{vmatrix}}{\begin{vmatrix} p_{2x_2} & p_{2x_3} & \cdots & p_{2x_M} \\ p_{3x_2} & p_{3x_3} & \cdots & p_{3x_M} \\ \vdots & \vdots & & \vdots \\ p_{Mx_2} & p_{Mx_3} & \cdots & p_{Mx_M} \end{vmatrix}} \theta_i \quad (49)$$

$$i = 2, 3, \dots, M.$$

In the solution (47), $\{p_2, \dots, p_M, \tilde{u}_{M+1}, \dots, \tilde{u}_N\}$ are all arbitrary functions of $\{x_2, x_3, \dots, x_N\}$ and every q_j is determined by a heat conductive equation

$$\lambda \theta_1 q_{jyy} - q_{jt} = 0, \quad j = 1, 2, \dots, M. \quad (50)$$

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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